

A5P1

October 4, 2025

```
[9]: import matplotlib.pyplot as plt
      from sklearn.metrics import roc_curve, auc, roc_auc_score
```

```
[11]: gtclabel = [1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]
      pprobscore = [0.85, 0.3, 0.35, 0.6, 0.55, 0.65, 0.4, 0.5, 0.45, 0.55, 0.75, 0.
      ↪85, 0.95, 0.1, 0.2, 0.15, 0.9, 0.7, 0.8, 0.3]
      fpr, tpr, thresholds = roc_curve(gtclabel, pprobscore)
      auc = roc_auc_score(gtclabel, pprobscore)
      plt.figure()
      plt.plot(fpr, tpr, label='ROC curve (area = %0.2f)' % auc)
      plt.plot([0, 1], [0, 1], linestyle='--', label='Random')
      plt.xlim([0.0, 1.0])
      plt.ylim([0.0, 1.0])
      plt.xlabel('False Positive Rate')
      plt.ylabel('True Positive Rate')
      plt.title('ROC Figure')
      plt.legend()
      plt.show()
```

